# Senior Smart Contract & Blockchain Protocol Engineer

**Company:** Copia Group
**Location:** Remote / International
**Employment Type:** Full-time
**Seniority:** Senior
**Compensation:** Competitive base salary + equity/token incentives

## Company Overview

Copia Group is building the infrastructure for the next generation of **Web3 financial products and payments**.

Our long-term vision is to build a **"Stripe for Web3"** — a scalable fintech and payments infrastructure layer that makes blockchain-based financial products easier to build, integrate, operate, and use.

As we expand internationally, including our growing presence in **Australia**, we are building an engineering team capable of creating secure, scalable, production-grade Web3 infrastructure.

This is a high-ownership environment where senior engineers will have meaningful influence over **protocol architecture, smart-contract design, security standards, and technical direction**.

## About the Role

We are looking for a **Senior Smart Contract & Blockchain Protocol Engineer** to design, develop, test, and optimize the smart-contract and protocol infrastructure behind Copia Group's Web3 fintech and payments platform.

You will work closely with backend, frontend, infrastructure, product, and security engineers to turn financial and payment requirements into **secure, efficient, and scalable on-chain systems**.

This role is particularly suited to an engineer who understands not only how to write Solidity, but also **why a protocol should be architected a certain way**.

## What You Will Do

- Design and develop production-grade smart contracts and blockchain protocols.
- Architect secure and upgradeable EVM-based systems.
- Translate product and financial requirements into on-chain mechanisms.
- Develop token, payment, settlement, treasury, and financial infrastructure.
- Integrate smart contracts with backend services, wallets, and frontend applications.
- Design efficient transaction and execution flows.
- Optimize smart contracts for gas efficiency and scalability.
- Develop robust access-control, authorization, and upgrade mechanisms.
- Design contract interfaces and modular protocol architectures.
- Build comprehensive automated testing using unit, integration, fuzz, invariant, and fork testing.
- Identify and remediate smart-contract security vulnerabilities.
- Participate in security reviews and external audit preparation.
- Investigate production issues across smart contracts, wallets, backend systems, and blockchain infrastructure.
- Evaluate new EVM standards, protocols, and blockchain technologies.
- Collaborate with product and engineering leadership on protocol strategy.
- Review code and mentor other blockchain engineers.
- Establish engineering standards for smart-contract development and deployment.
- Contribute to technical documentation and protocol specifications.

## Core Technical Responsibilities

### Smart Contract Engineering

- Solidity development
- EVM architecture
- Contract upgradeability
- Modular contract design
- Access control
- Token standards
- Payment and settlement mechanisms
- Treasury systems
- Contract interaction patterns
- Gas optimization
- Security-oriented development

### Protocol Engineering

- Design decentralized protocols and financial primitives.
- Define state transitions and protocol invariants.

- Design permission and authorization models.
- Build composable protocol components.
- Consider economic and technical attack vectors.
- Design reliable on-chain/off-chain execution flows.
- Evaluate trade-offs between decentralization, security, scalability, and UX.

### Testing & Security

You will be expected to treat security as a core engineering responsibility.

Experience with some of the following is highly valued:

- Foundry
- Hardhat
- Forge
- Fuzz testing
- Invariant testing
- Property-based testing
- Fork testing
- Static analysis
- Formal verification
- Certora
- Slither
- Security reviews
- Smart-contract audits
- Threat modeling

## Technical Environment

Our stack may include:

- **Solidity**
- **EVM**
- Ethereum
- Layer 2 networks
- Foundry
- Hardhat
- OpenZeppelin
- Ethers.js
- Viem
- Web3.js
- The Graph
- TypeScript
- Node.js

- Rust
- PostgreSQL
- Redis
- Docker
- Kubernetes
- CI/CD

The exact technology stack will evolve with the protocol and product roadmap.

## Required Qualifications

- **4+ years of professional blockchain/Web3 engineering experience**, or equivalent depth of smart-contract development experience.
- Strong professional experience with **Solidity and EVM development**.
- Demonstrated experience designing and deploying production smart contracts.
- Strong understanding of Ethereum and EVM execution.
- Experience with smart-contract testing and security practices.
- Strong understanding of blockchain transaction lifecycle and gas mechanics.
- Experience integrating smart contracts with backend and/or frontend applications.
- Ability to reason about protocol architecture and security.
- Strong software engineering fundamentals.
- Experience with Git, code review, testing, and CI/CD workflows.
- Strong written and verbal communication skills.

## Strongly Preferred

- Experience building **DeFi, payments, stablecoin, wallet, or financial protocols**.
- Experience with ERC-20, ERC-721, ERC-1155, ERC-4626 or related standards.
- Experience with account abstraction and **ERC-4337**.
- Experience with EIP-7702 or emerging account-abstraction standards.
- Experience with Layer 2 networks.
- Experience with cross-chain protocols or interoperability.
- Rust experience.
- Experience with Foundry.
- Experience preparing protocols for external security audits.
- Experience responding to audit findings.
- Experience with formal verification or Certora.
- Experience building blockchain indexers or subgraphs.
- Previous startup or founding-engineer experience.

## What Would Make You Exceptional

You don't simply write smart contracts — **you think like a protocol architect**.

You can take a high-level financial or payment requirement and break it down into:

**business logic → protocol design → state transitions → security model → smart contracts → testing → deployment → application integration.**

You understand that production blockchain engineering requires more than Solidity. You consider:

- Security
- Economic incentives
- Gas costs
- Upgradeability
- Failure scenarios
- Transaction ordering
- Reentrancy and authorization
- Oracle and external dependency risks
- Cross-contract interactions
- User experience
- Off-chain/on-chain boundaries

You are comfortable taking ownership of a protocol component from **architecture through production**.

# Why Join Copia Group?

- Build core infrastructure for a **“Stripe for Web3”** fintech/payments platform.
- Take ownership of critical protocol architecture.
- Work on real-world Web3 financial and payment infrastructure.
- Influence smart-contract and blockchain technology decisions from an early stage.
- Work closely with product, backend, frontend, infrastructure, and security teams.
- International and remote-first working environment.
- Opportunity to work across **Web3, fintech, payments, DeFi, stablecoins, and blockchain infrastructure**.
- Competitive compensation with potential **equity/token upside**.
- Opportunity to grow into a Principal Engineer, Protocol Lead, or technical leadership role.

# Compensation

**Indicative base salary: $200,000–$250,000 gross per year**

For exceptional candidates with deep protocol, DeFi, security, or financial infrastructure experience, the total compensation package may include:

- Competitive base salary

- Equity
- Token incentives where applicable
- Performance-based bonus
- Location-specific benefits

The final package will depend on experience, location, technical depth, and level of ownership.

## Hiring Process

1. Introductory conversation with the recruitment team
2. Technical interview focused on Solidity, EVM, protocol architecture, and security
3. Practical technical/product assessment
4. Architecture and protocol discussion with technical leadership
5. Final interview
6. Offer

**We're looking for an engineer who wants to build secure financial infrastructure on-chain — not simply write smart contracts.**